

TOLUWANIMI ODUNEWU

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RESEARCH INTERESTS

- Drug-response and molecular property prediction
- Evaluation and trustworthiness of predictive models in pharmacology
- Generative and machine-learning models for drug discovery
- Computational pharmacology and cheminformatics
- Multi-agent systems for scientific automation

EDUCATION

BSc (Hons) University of Lagos, Pharmacology, Therapeutics and Toxicology

January 2026

Grade: 4.47/5.0 CGPA, Second Class Honours (Upper Division, 2:1). Final-year performance: 16 A grades and 2 B grades across 18 courses. *University of Lagos First Class threshold: 4.50/5.00.*

Dissertation: Evaluation of the Neuroprotective and Neurotoxic effects of selected Anti-seizure medications using the *Drosophila melanogaster* model

Advisor: Professor Ismail Ishola

Relevant Coursework (grade): Immunopharmacology: Pain, Inflammation, Anti-Inflammatory and Anti-Allergic Agents, and Anaphylaxis (A); Quantitative Pharmacology (A); Medical Statistics for Pharmacology (A); Pharmacokinetics and Pharmacogenetics (A); Research Methodology and Scientific Writing (A); Neuropharmacology (A); Pharmacoepidemiology (A); Systemic Toxicology (B); Research Project (A).

TECHNICAL SKILLS

Programming & AI Development

Python, JavaScript/TypeScript, FastAPI, Next.js 14, React, Node.js, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Prompt Engineering, Vector Embeddings, Semantic Search, Document Processing

Machine Learning & Data Science

PyTorch, RDKit, Chemprop, ChemBERTa, XGBoost, Random Forest, Scikit-learn, ADMET Prediction, Molecular Property Calculation, Pandas, NumPy, Matplotlib | Model Evaluation, Benchmark Design, Bootstrap and Wilson Confidence Intervals

Computational Chemistry

Schrödinger Suite, Molecular Docking, MM-GBSA, Virtual Screening, SMILES/SMARTS/SMIRKS notation, Bemis-Murcko Scaffolds, Molecular Structure Analysis, Drug-Drug Interaction Prediction

Laboratory & Research

In-vivo models (*Drosophila melanogaster*), Experimental Design, Data Analysis, Literature Review, Scientific Writing, PubMed, Google Scholar, Academic Databases

GitHub

<https://github.com/Nimieeee>

RESEARCH & PROJECT EXPERIENCE

Allosteric Inhibition of PBP2a in Methicillin-Resistant *Staphylococcus aureus*

2026

AiSynth Labs · in progress

- Structure-based analysis of the allosteric mechanism from PDB 4CJN: measured the quinazolinone allosteric site at **58.3 Å** from the Ser403 catalytic nucleophile, mapped the salt-bridge network linking the two lobes, and located the clinical resistance mutations **N146K** and **E150K** directly on the allosteric trigger.
- Found that geometric druggability scoring did not identify the functional site: the highest-scoring predicted pocket (0.636) lay 29.7 Å from the crystallographic ligand with no contact-residue overlap, while the pocket that recovered 3 of 12 true contact residues scored lowest (0.228). A reminder that a cavity's druggability score describes its shape, not its function.
- Generated 47 candidate allosteric binders and passed them through a defence-in-depth gate cascade (validity, PAINS and Brenk alerts, ADMET thresholds, QED), 38 surviving. Chemical-space analysis against the crystallographic ligand showed maximum Tanimoto 0.191 and Fsp³ 0.50 against 0.00, i.e. a genuinely distinct, more three-dimensional series rather than close analogues.

Data-Efficiency Crossover: Fine-Tuning Chemistry Foundation Models vs. Classical ML

2026

Personal Project

Link: <https://github.com/Nimieeee/chemberta-vs-rf-data-efficiency>

- Designed and ran a controlled learning-curve study fine-tuning DeepChem's ChemBERTa-77M-MLM (a pretrained transformer, 77M PubChem molecules) against Random Forest on ECFP4 fingerprints across four real TDC ADMET endpoints (aqueous solubility, lipophilicity, acute toxicity, hERG cardiotoxicity).
- Found fine-tuning decisively outperforms classical baselines with no data-efficiency crossover on two endpoints (test R^2 0.41→0.73 on solubility, 0.39→0.52 on lipophilicity), while identifying, via target-distribution analysis, an endpoint where it does not.
- Diagnosed and corrected a methodological confound (unstandardized regression targets across endpoints of differing label variance) that had produced a misleading result, then re-validated the full study with matched target scaling, matched class weighting, and bootstrap 95% confidence intervals on every reported score.
- Published complete methodology, results, and code publicly on GitHub.

Lead Developer (Benchside formerly PharmGPT)

2026

Personal Project

Link: <https://benchside.vercel.app/>

- Architected and launched a browser-based, agentic drug-discovery platform spanning ADMET prediction, lead optimization, and autonomous literature research.
- Lead optimization: an LLM-governed agent that selects from a 479-entry SMIRKS transformation library, with RDKit executing the reactions to generate analogs ranked by multi-objective ADMET and synthetic accessibility.
- ADMET scoring: a local Chemprop v2 engine scoring 54 endpoints against DrugBank-approved percentile distributions.
- Deep research: a four-node agentic pipeline (Planner, Researcher, Reviewer, Writer) that synthesizes literature with verified citations and live RCSB PDB structure lookups.
- Benchmarked clinical reasoning on the NAPLEX licensing exam, reaching 73% accuracy ahead of a leading frontier reasoning model, and reduced hallucination from 29% to 21% through domain calibration.
- Deployed full-stack: Next.js (Vercel) frontend, FastAPI (PM2, Nginx) backend, Supabase/PostgreSQL, and multi-provider LLM routing.

A Vision-Language Agentic Pipeline for Lead Optimization: Defence-in-Depth Chemistry Gates and Cross-Provider Model Evaluation

2026

Single-Author Research Preprint · ChemRxiv

Link: <https://doi.org/10.26434/chemrxiv.15004729/v1>

- Designed and ran a five-part quantitative evaluation of Benchside's twelve-stage lead-optimization pipeline: an unbiased ChEMBL-37-derived matched-molecular-pair scale-up (2,000 pairs, 15.5% exact transformation recovery, 95% CI 14.0–17.2, against 53.3% on a curated set), a 129-marketed-drug scaffold-preservation audit, and an end-to-end case study across two contrasting drug targets.
- Introduced and validated a "defence-in-depth" architecture, deterministic chemistry gates wrapped around each language-model agent stage, showing the soft Murcko scaffold gate raises scaffold preservation to 100% while roughly halving the analog set, and that vision-language classifications reach perfect self-consistency (Jaccard 1.00 across 28 pairwise comparisons) once gated.
- Benchmarked fourteen vision-capable and eleven text-based language models across the pipeline's three agent stages for structured-output reliability, producing measured, not asserted, recommendations for production model selection.
- Released full code, data, and provenance-manifested raw outputs open source (MIT licence) for reproducibility.

Schrödinger Suite MCP Server

2026

Personal Project

Link: <https://github.com/Nimieeee/schrodinger-mcp>

- Built a Model Context Protocol server exposing Schrödinger Suites 2026 drug-discovery workflows as agentic tools, letting an AI assistant run PDB retrieval, protein and ligand preparation, Glide docking (SP and XP), QikProp ADMET, Prime MM-GBSA, SiteMap, and Jaguar QM end to end and return ranked results with files.
- Engineered an asynchronous job-orchestration layer (detached supervisor, status polling, job registry) in Python with a regression test suite, isolating the server's dependencies from Schrödinger's bundled runtime.

FutureGRIN NextGen Training Program

2025

Computational Research Trainee

Advisor: Adediji Ayomide

- Applied skills from the training program to a team-based computational drug discovery project.
- Performed structure-based virtual screening of the LOTUS natural products database by running molecular docking simulations with the Schrödinger Suite.
- Analyzed docking results to evaluate binding affinities and prioritize lead compounds based on their predicted interactions within the DYRK1A active site.

Undergraduate Researcher & Lab Assistant

Advisor: Professor Ismail Ishola

Title: Evaluation of the Neuroprotective and Neurotoxic effects of selected Anti-seizure medications using the *Drosophila melanogaster* model

- Investigated the cellular and behavioral impacts of anticonvulsant drugs in a living biological system.
- Designed and executed experimental protocols, collected quantitative data, and performed statistical analysis to assess therapeutic and adverse effects.
- Managed the husbandry and daily maintenance of the *Drosophila melanogaster* laboratory, ensuring the health and viability of all genetic stocks.

LEADERSHIP & PROFESSIONAL EXPERIENCE

AiSynth Labs

November 2025 – Present

Founder & Lead Scientist

- Founded an initiative to democratize computer-aided drug discovery, running training bootcamps that teach CADD to students and researchers; launched the first cohort in 2026.
- Lead an in-house research program, including "Overcoming the Beta-Lactam Barrier: Computational Identification of Candidate Allosteric PBP2a Binders for MRSA," applying structure-based and computational methods to antimicrobial resistance.

Sephar Innovations

March 2026 – Present

AI Engineer

- Build AI features for Xepho (xepho.io), a learning-management platform, generating VARK-aligned learning content: automated slides, explainer videos, and podcast-style audio.
- Engineer LLM-driven pipelines that transform source material into multi-modal study formats.

PUBLICATIONS

Preprint: Odunewu, T. (2026). *A Vision-Language Agentic Pipeline for Lead Optimization: Defence-in-Depth Chemistry Gates and Cross-Provider Model Evaluation*. ChemRxiv. <https://doi.org/10.26434/chemrxiv.15004729/v1> (single author).

Under review: Odunewu, T. (2026). *Defence-in-Depth Is Not Coverage: A Failure Analysis of an Agentic Lead-Optimization Pipeline*. AI for Drug Discovery workshop, NeurIPS 2026.

Under review: Odunewu, T. S., Samuel, F. J., Ogunnoiki, A. O., & Ishola, I. O. (2026). *Multi-Generational Neurodevelopmental and Cognitive Toxicity of Antiseizure Medications in Drosophila melanogaster*. Discover Toxicology (Springer Nature), in peer review. **(first author)**

CONFERENCES & PRESENTATIONS

Poster Presentation: *Alzheimer's Association International Conference (AAIC)*, 2026. "Targeting Hyperexcitability & Oxidative Stress In Dementia: Comparative Cognition-sparing Anti-seizure Strategies"

Accepted: *NISPET 2026*, Nigerian Society for Pharmacological Sciences and Experimental Therapeutics, 2026. "A Vision-Language Agentic Pipeline for Lead Optimization: Defence-in-Depth Chemistry Gates and Cross-Provider Model Evaluation"

Abstract submitted: *Pharmacology 2026*, British Pharmacological Society, Manchester, December 2026. "Deterministic chemistry gates constrain vision-language model error in an agentic lead-optimisation pipeline, but benchmark construction governs reported performance"

PROFESSIONAL AFFILIATIONS

Member, International Society to Advance Alzheimer's Research and Treatment (ISTAART), 2025–Present

LANGUAGES

English: Native Language · **Yoruba:** Native Language